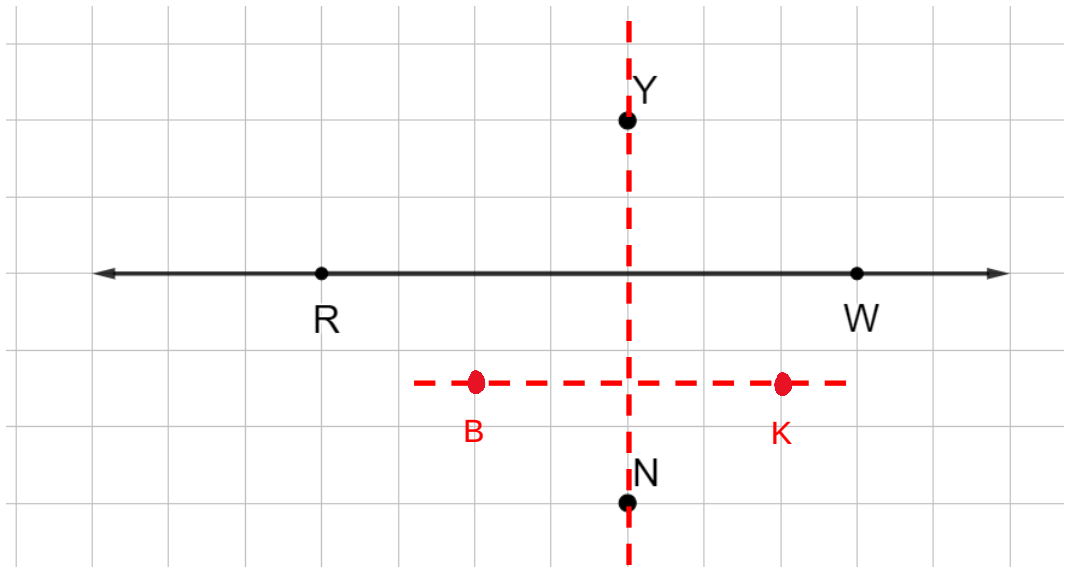


Line RW and points N and Y are shown.



1. Fold the paper so that the fold goes through points N and Y and line RW coincides with itself. Draw a dashed line on the grid that shows the fold.
2. Two students wrote a description of the relationship the dashed line has with line RW .

Olivia	Makayla
The dashed line divides line RW in half.	The dashed line makes a right angle with line RW .

Write a brief explanation that describes which student's description is not correct.

Olivia's description is not correct; the dashed line is 4 units from R and 3 units from W .

3. Fold the dashed line on itself so that point N coincides with line RW . Draw a line along the fold and name the line BK .

Complete the statements about the construction of the dashed line and line BK using words from the word bank.

Word Bank

line

line segment

parallel

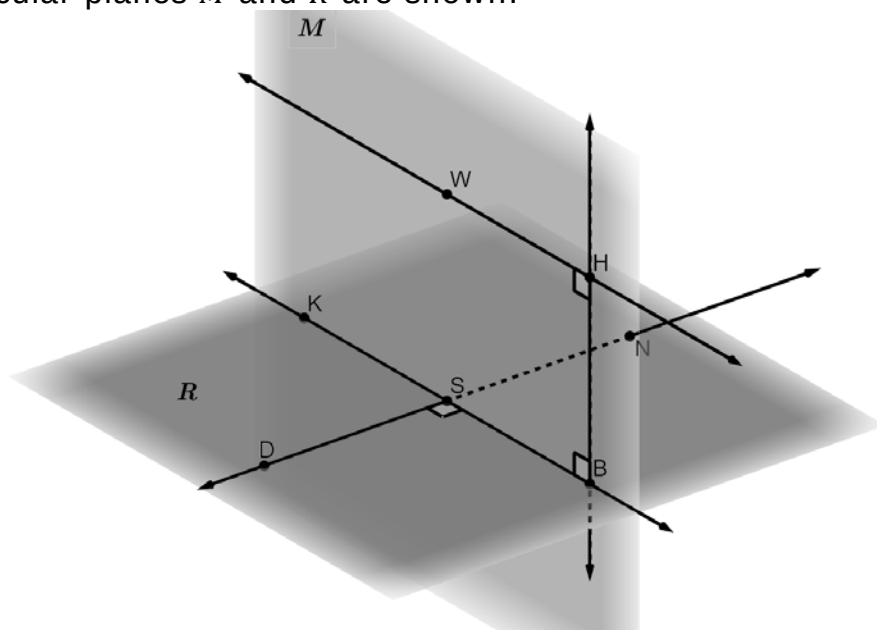
perpendicular

plane

point

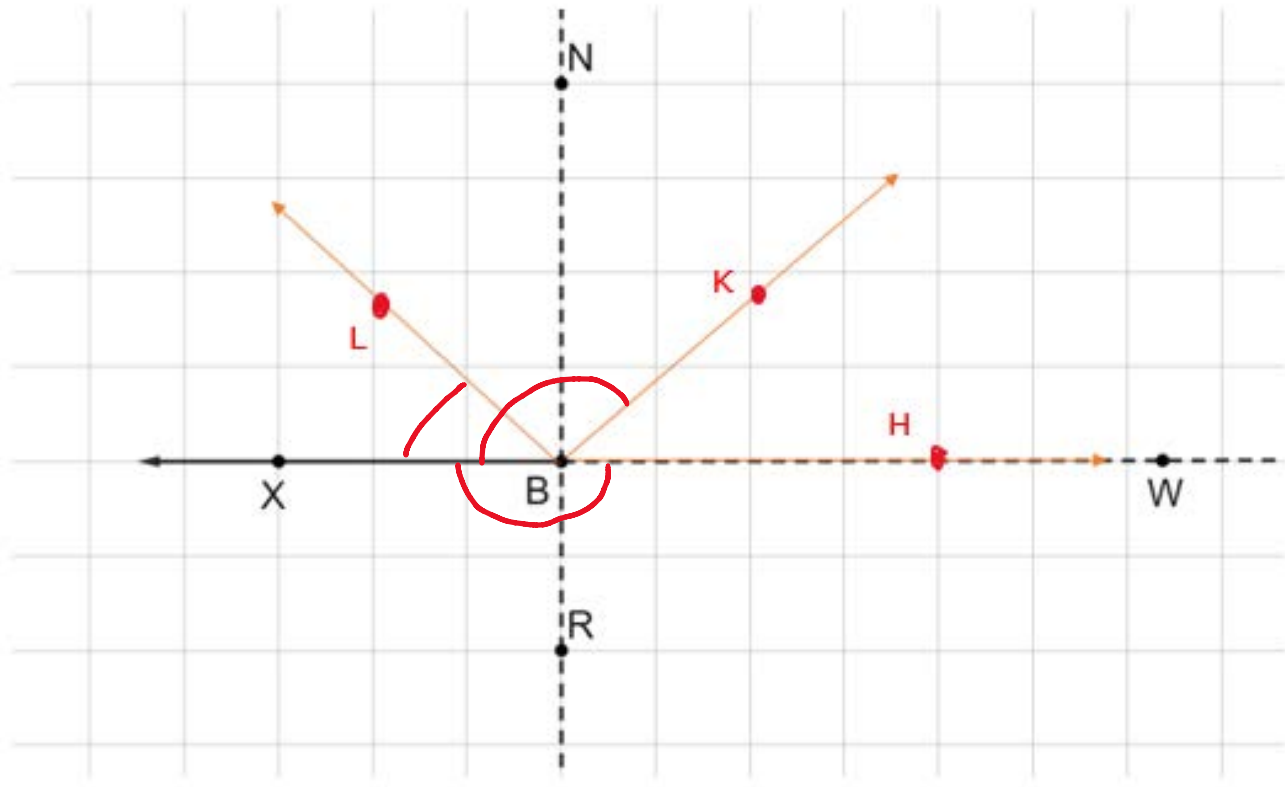
4. The homework paper represents a plane.
5. Lines RW and BK are parallel.
6. The dashed line that passes through points N and Y is perpendicular to lines RW and BK .

Perpendicular planes M and R are shown.



7. Identify two lines that are parallel.
 $\overrightarrow{WH} \parallel \overrightarrow{KS}$
8. Identify two lines that are perpendicular and lay in plane R .
 $\overrightarrow{KS} \perp \overrightarrow{DS}$
9. Identify two lines that are perpendicular and lay in plane M .
 $\overrightarrow{WH} \perp \overrightarrow{HB}$
10. Identify two lines that are skew. $\overrightarrow{HB}$ and $\overrightarrow{DS}$ or $\overrightarrow{HW}$ and $\overrightarrow{DS}$

Lines RN , BW , and ray BX are shown on grid paper.



1. Suppose you have a piece of patty paper that has ray BK on it. Describe how ray BK can be used to create angle XBK such that angle XBK is obtuse.

Answers may vary. Possible answer: $\overrightarrow{BK}$ could lie between $\overrightarrow{BN}$ and $\overrightarrow{BW}$.

2. Draw angle XBK on the grid paper.

Answers may vary, but should match #1.

3. Suppose you have a piece of patty paper that has ray BH on it. Describe how ray BH can be used to create angle XBH such that angle XBH is a straight angle.

$\overrightarrow{BH}$ should lie on $\overrightarrow{BW}$.

4. Draw angle XBH on the grid paper.

5. Suppose you have a piece of patty paper that has ray BL on it. Describe how ray BL can be used to create angle XBL such that angle XBL is an acute angle.

Answers may vary. Possible answer: $\overrightarrow{BL}$ could lie between $\overrightarrow{BX}$ and $\overrightarrow{BN}$.

6. Draw angle XBL on the grid paper.
Answers may vary, but should match #5.

Micah listed key characteristics of points, lines, and planes. His list is shown.

Point	Line	Plane
has no dimension	has infinite length with no width or height	has three dimensions
indicates a position or location	has a measure of 180°	has infinite length and width, but no height
goes on forever	goes on forever in both directions	has a rectangular shape when drawn
can vary in size	can be named using two capital letters or one lowercase letter	goes on forever

Micah's list of characteristics has a few mistakes.

7. Which, if any, of Micah's characteristics of a point is not correct?

"Goes on forever" and "can vary in size" are not correct because a point has no dimension and no size.

8. Which, if any, of Micah's characteristics of a line is not correct?

All are correct.

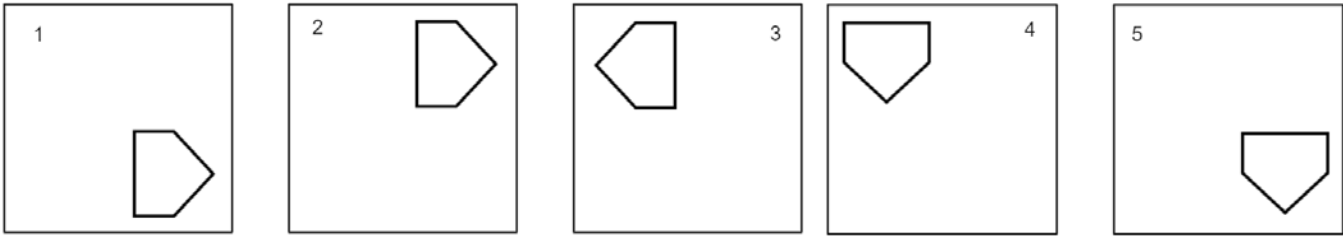
9. Which, if any, of Micah's characteristics of a plane is not correct?

"Has 3 dimensions" is not correct because a plane has two dimensions.

10. Select all of the terms from the word bank that are considered undefined terms in Geometry.

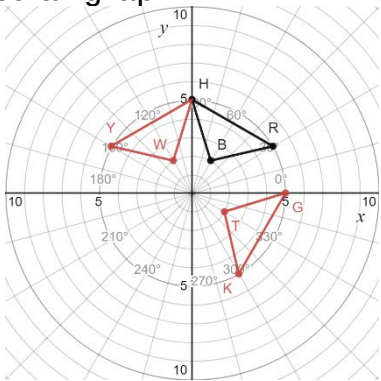
Word Bank					
line	line segment	parallel	perpendicular	plane	point

Five frames that feature a polygon in different positions are shown.



	Movement from...	Description
1.	Frame 1 to Frame 2	Translation
2.	Frame 2 to Frame 3	Reflection
3.	Frame 3 to Frame 4	Rotation
4.	Frame 4 to Frame 5	Translation

Triangle RHB is shown on the circular graph.



5. Plot each of the points.

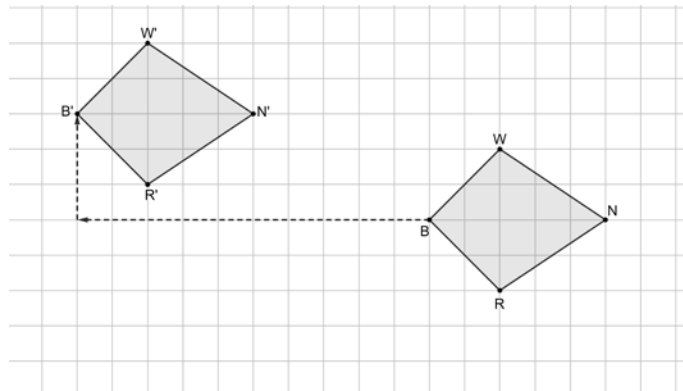
Point	Description
W	Plotted on the 120° line and on the same circle as B .
Y	Plotted on the 150° line and on the same circle as R .
K	Plotted on the 300° line and on the same circle as H .
T	Plotted on the 330° line and on the same circle as B .
G	Plotted on the 360° line and on the same circle as R .

6. Complete the statements about the triangles.

$\triangle HYW$ is a ☒ reflection
☐ rotation of $\triangle HRB$.

$\triangle GKT$ is a ☐ reflection
☒ rotation of $\triangle HRB$.

A diagram of $WNRB$ and $W'N'R'B'$ is shown. Arrows are drawn from vertex B to vertex B' .



Complete the statements.

7. The arrows show a possible path of movement. This path is called a

- ☐ reflection.
- ☐ rotation.
- ☒ translation.

8. The same path ☒ exists
☐ does not exist for all points on a diagram.

9. The path is called ☐ isometry.
☒ a mapping.

10. The transformation shown on the grid can be referred to as

- ☒ an isometry
- ☐ a mapping
- ☐ a rigid motion

or as

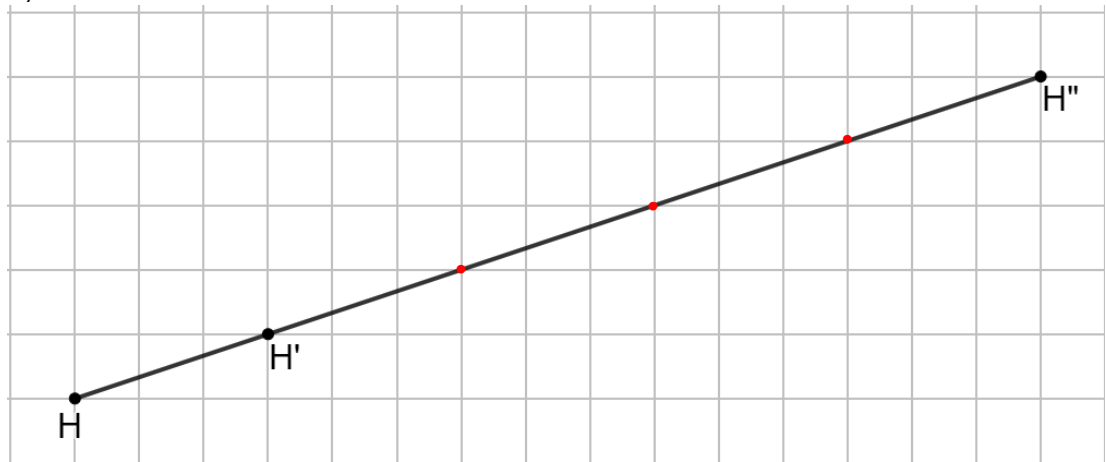
- ☐ an isometry
- ☐ a mapping
- ☒ a rigid motion

because

the transformation

- ☐ preserved distance.
- ☐ preserved angle measure.
- ☒ preserved angle measure and distance.

Points H , H' , and H'' are shown.



1. Determine the number of copies of $\overline{HH'}$ that will fit on $\overline{HH''}$.
5
2. Use the definition of dilation to explain why $\overline{HH''}$ is a dilation of $\overline{HH'}$.
 $\overline{HH''}$ is 5 times the length of $\overline{HH'}$.
3. Complete the statement.

Point

☐ H
☒ H'
☐ H''

 partitions

☐ $\overline{HH'}$
☐ $\overline{H'H''}$
☒ $\overline{HH''}$

 so that the ratio of $\frac{HH'}{HH''}$ is

☐ $\frac{1}{4}$
☒ $\frac{1}{5}$

Use the Pythagorean theorem to complete the table.

	Segment	Length
4.	$\overline{HH'}$	$1^2 + 3^2 = c^2$ $10 = c^2$ $c = \sqrt{10}$
5.	$\overline{H'H''}$	$4^2 + 12^2 = c^2$ $160 = c^2$ $c = 4\sqrt{10}$
6.	$\overline{HH''}$	$5^2 + 15^2 = c^2$ $250 = c^2$ $c = 5\sqrt{10}$

7. Complete the statements.

Point H' is

☐

☒

$\frac{1}{4}$
 $\frac{1}{5}$

 of the distance from H to H'' . The scale factor that can be

used to multiply

☐

☐

☒

$\overline{HH'}$
 $\overline{H'H''}$
 $\overline{HH''}$

 by to produce

☒

☐

☐

$\overline{HH'}$
 $\overline{H'H''}$
 $\overline{HH''}$

 is

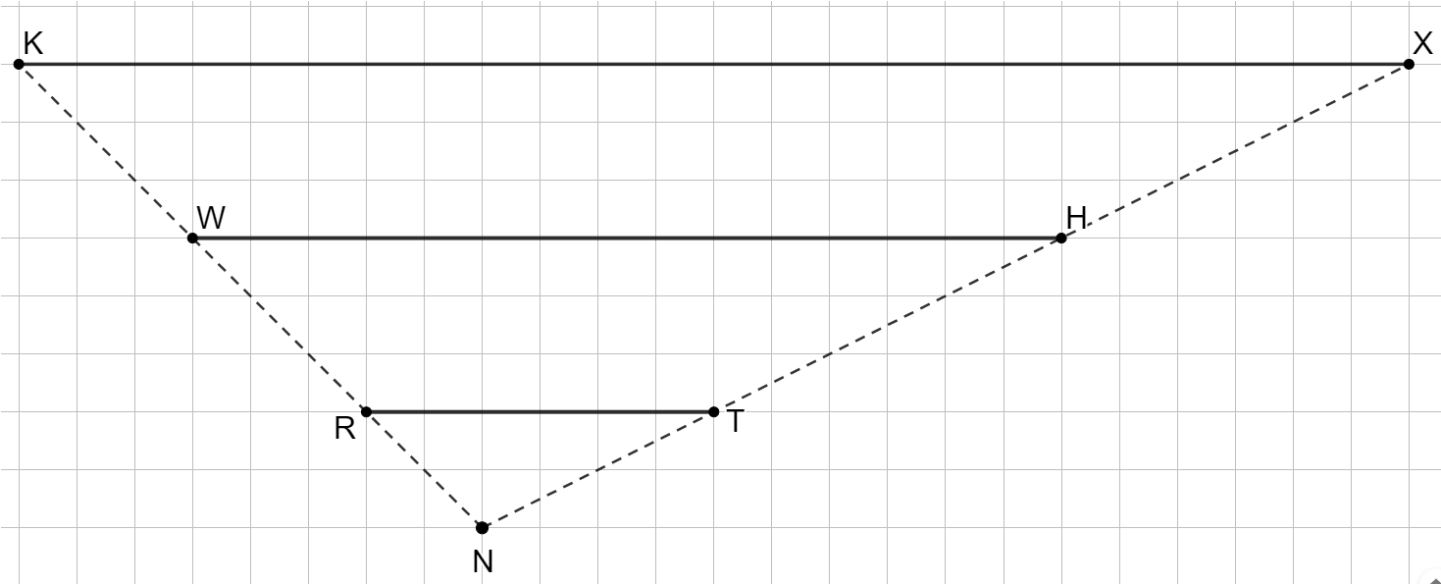
☐

☒

$\frac{1}{4}$
 $\frac{1}{5}$

.

Line segments KX , WH , and RT are shown where $\overline{RT}$ is dilated using point N to create $\overline{WH}$ and $\overline{KX}$.



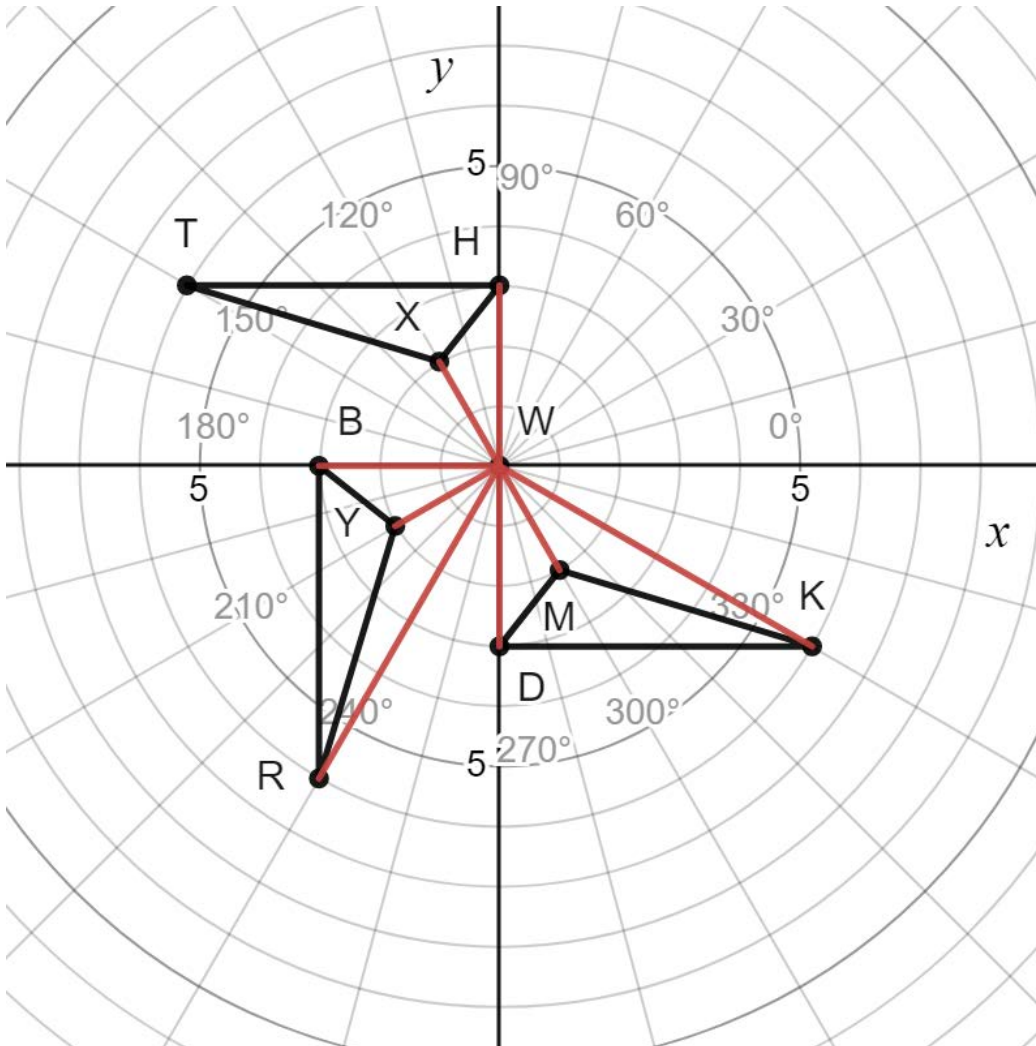
Determine each of the ratios.

	Ratio	Length
8.	$NR:NW$	$2\sqrt{2}:5\sqrt{2}$
9.	$RT:WH$	$6:15$
10.	$RT:KX$	$6:24$

Geometry
Unit 1
Lesson 5 Homework

Name Answer Key
Date _____
Period _____

Triangles THX , YBR , and DKM are shown on the circular grid.



Draw each of the angles on the grid.

- | | |
|-----------------|-----------------|
| 1. $\angle XWY$ | 4. $\angle BWD$ |
| 2. $\angle KWR$ | 5. $\angle BWH$ |
| 3. $\angle DWH$ | 6. $\angle MWX$ |

7. Select all of the angles drawn on the grid whose measure is 90° .

- ☒ $\angle XWY$
- ☒ $\angle KWR$
- ☐ $\angle DWH$
- ☒ $\angle BWD$
- ☒ $\angle BWH$
- ☐ $\angle MWX$

8. Select all of the angles drawn on the grid whose measure is 180° .

- ☐ $\angle XWY$
- ☐ $\angle KWR$
- ☒ $\angle DWH$
- ☐ $\angle BWD$
- ☐ $\angle BWH$
- ☒ $\angle MWX$

Complete the statements.

9. $\triangle THX$ is created by rotating

☒ $\triangle KDM$
☐ $\triangle RBY$

 180° clockwise about point W .

10. $\triangle RBY$ is created by rotating

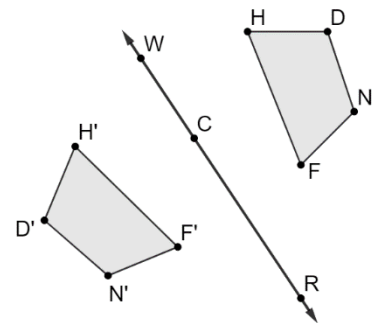
☒ $\triangle KDM$
☐ $\triangle THX$

 90° clockwise about point W .

Quadrilaterals $HDNF$ and $H'D'N'F'$ are shown.

1. Complete the table to describe the single transformation that was used to create $H'D'N'F'$.

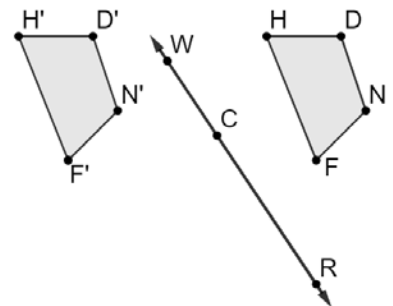
Type	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☒ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☒ Line of reflection is $\overleftrightarrow{WR}$. ☐ Down ☐ Left ☐ Right ☐ Up	



Quadrilaterals $HDNF$ and $H'D'N'F'$ are shown.

2. Complete the table to describe the single transformation that was used to create $H'D'N'F'$.

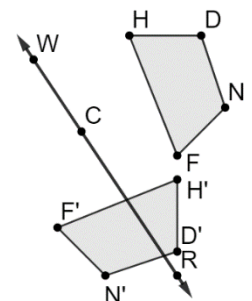
Type	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☒ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☒ Left ☐ Right ☐ Up	



Quadrilaterals $HDNF$ and $H'D'N'F'$ are shown.

3. Complete the table to describe the single transformation that was used to create $H'D'N'F'$.

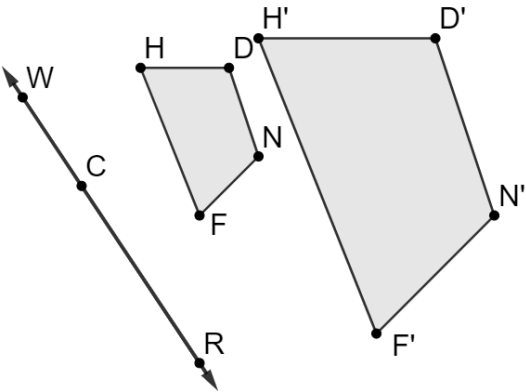
Type	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☒ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☒ Center of rotation is Point C. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	90° clockwise or 270° counterclockwise



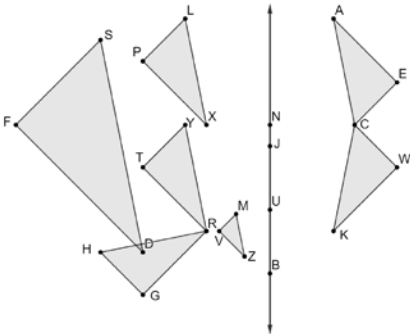
Quadrilaterals $HDNF$ and $H'D'N'F'$ are shown.

4. Complete the table to describe the single transformation that was used to create $H'D'N'F'$.

Type	Center, Line, or Movement	Degrees or Scale Factor
☒ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☒ Center of dilation is Point W . ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	2



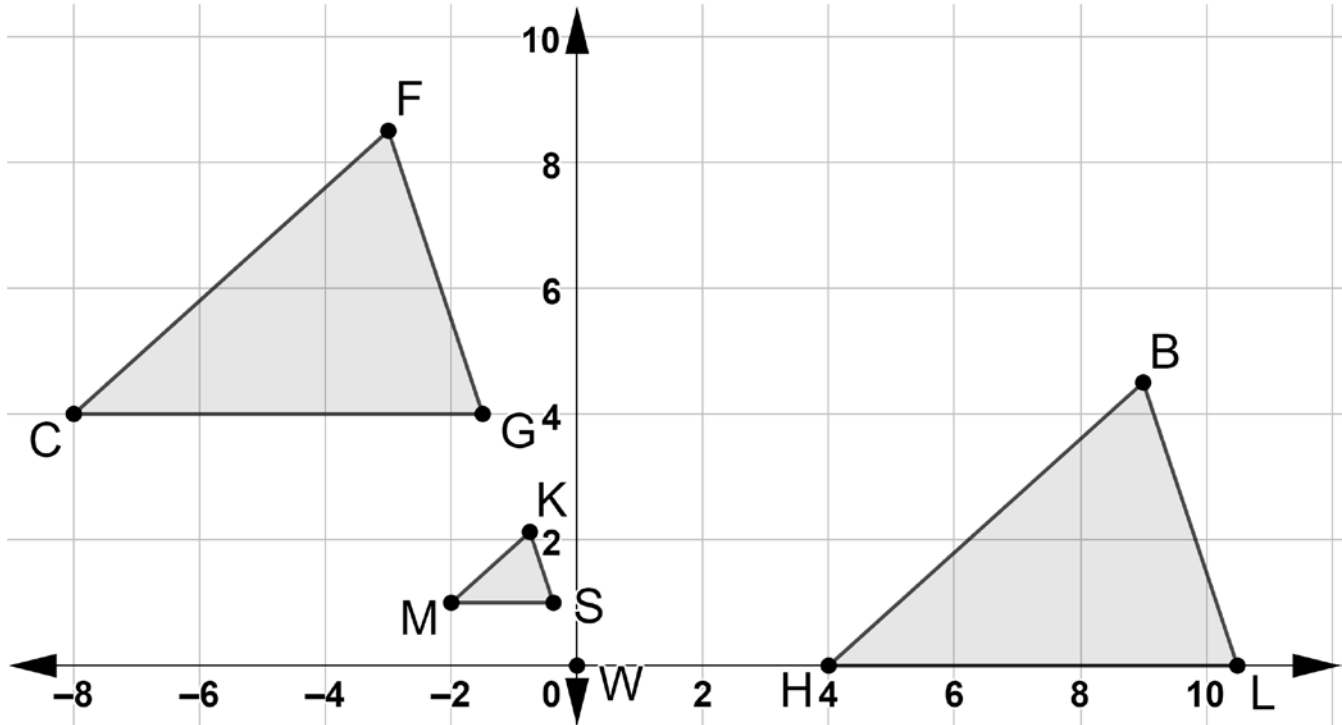
The diagram shows six different single transformations of $\triangle YTR$.



Complete the table.

Triangle	Type of Transformation
5. $\triangle MVZ$	Dilation centered at Point B .
6. $\triangle HGR$	Rotation around Point R .
7. $\triangle CWK$	Reflection over Line NB .
8. $\triangle LPX$	Translation up.
9. $\triangle SFD$	Dilation centered at Point U .
10. $\triangle CEA$	Rotation 180° around Point N .

Three triangles are shown on the coordinate grid. $\triangle LBH$ and $\triangle SKM$ are created using a single transformation of $\triangle GFC$.



Complete each statement using an angle or a segment from $\triangle GFC$.

1. $\overline{LB} \cong \overline{GF}$

2. $\overline{SM} \sim \overline{GC}$

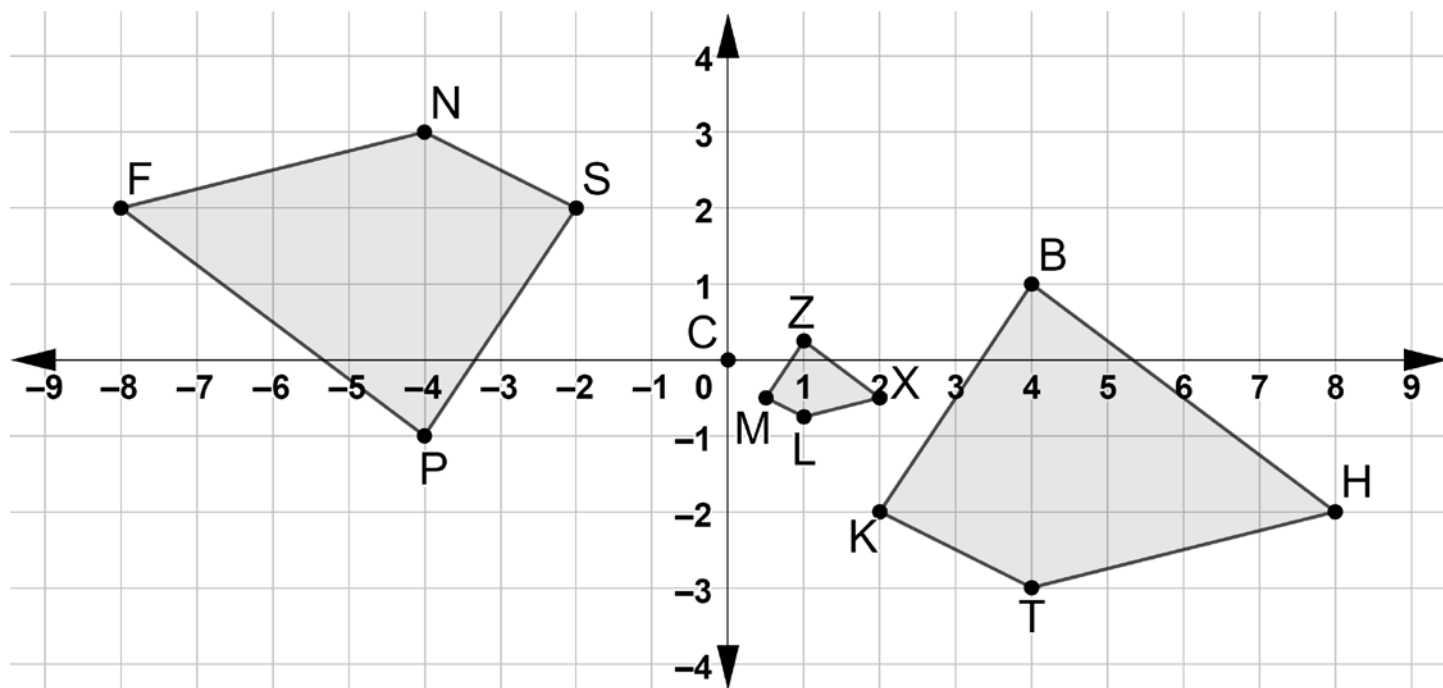
3. $\overline{BH} \cong \overline{FC}$

4. $\overline{LH} \cong \overline{GC}$

5. $\overline{MK} \sim \overline{CF}$

6. $\overline{SK} \sim \overline{GF}$

Three quadrilaterals are shown on the coordinate grid. Quadrilaterals $PFNS$ and $ZXLM$ are created using a single transformation of quadrilateral $BHTK$.



Complete the following statements with the $\cong$ or $\sim$ symbol.

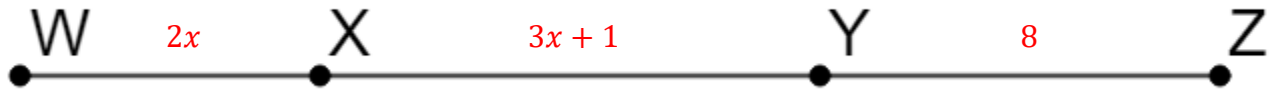
7. $\angle MLX \cong \angle KTH$

8. $\overline{HB} \cong \overline{PF}$

9. $\overline{ZM} \sim \overline{BK}$

10. $\angle BHT \cong \angle NFP$

Line segment WZ is shown where $WX = 2x$, $XY = 3x + 1$, $YZ = 8$, and $WZ = 29$.



1. Find the value of x .

$$\begin{aligned}2x + 3x + 1 + 8 &= 29 \\5x + 9 &= 29 \\5x &= 20 \\x &= 4\end{aligned}$$

2. Find the length of $\overline{WX}$.

$$\begin{aligned}\overline{WX} &= 2x \\&= 2 \cdot 4 \\&= 8\end{aligned}$$

3. Find the length of $\overline{XY}$.

$$\begin{aligned}\overline{XY} &= 3x + 1 \\&= 3 \cdot 4 + 1 \\&= 12 + 1 \\&= 13\end{aligned}$$

4. Find the length of $\overline{XZ}$.

$$\begin{aligned}\overline{XZ} &= \overline{XY} + \overline{YZ} \\&= 13 + 8 \\&= 21\end{aligned}$$

P , Q , and R are three collinear points that exist such that $\overline{PQ}$ and $\overline{QR}$ are congruent. $PQ = 6x - 8$ and $QR = 10x - 44$.

5. Draw a diagram that represents points P , Q , and R .



6. Find the value of x .

$$\begin{aligned} 6x - 8 &= 10x - 44 \\ 36 &= 4x \\ 9 &= x \end{aligned}$$

7. How long is segment $\overline{PQ}$?

$$\begin{aligned} 6x - 8 &= \overline{PQ} \\ &= 6 \cdot 9 - 8 \\ &= 46 \end{aligned}$$

8. How long is segment $\overline{QR}$?

$$\begin{aligned} 10x - 44 &= \overline{QR} \\ &= 10 \cdot 9 - 44 \\ &= 46 \end{aligned}$$

9. How long is segment $\overline{PR}$?

$$\begin{aligned} \overline{PQ} + \overline{QR} &= \overline{PR} \\ &= 46 + 46 \\ &= 92 \end{aligned}$$

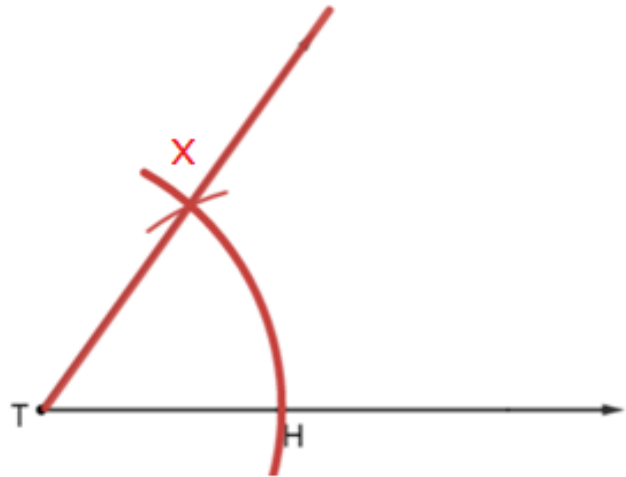
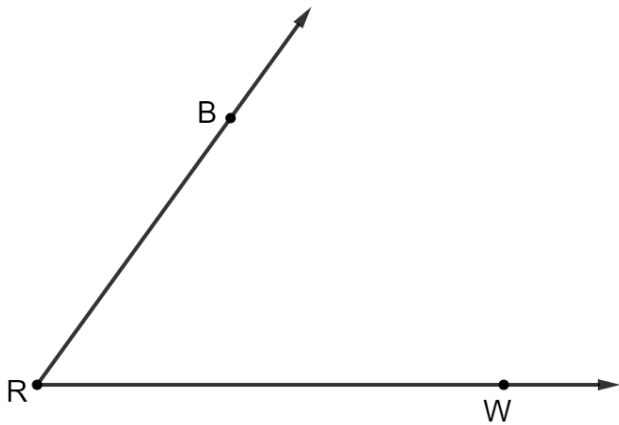
10. Use a compass and straightedge to make a copy of this segment.



Geometry
Unit 1
Lesson 9 Practice

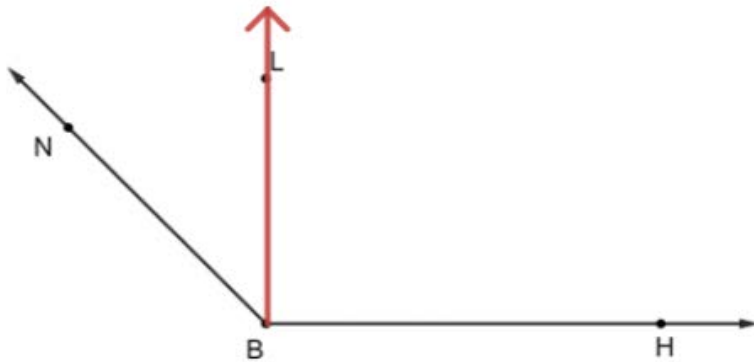
Name Answer Key
Date _____
Period _____

Angle BRW and ray TH are shown.



1. Use $\overrightarrow{TH}$ to create $\angle HTX$, a copy of $\angle BRW$.

Point L lies in the interior of angle NBH as shown.

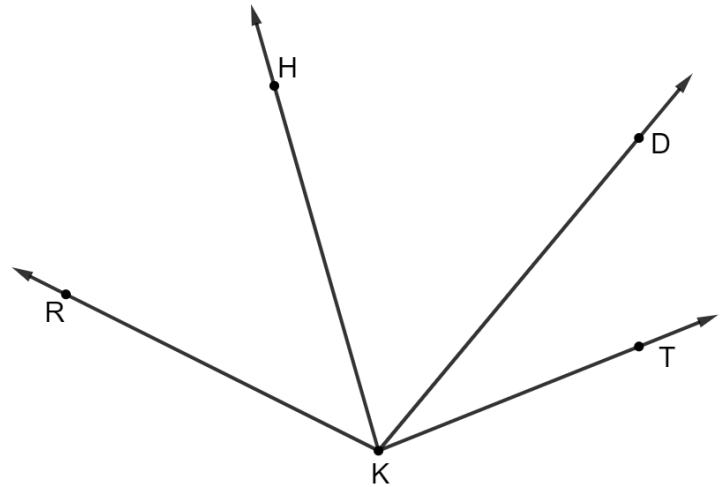


2. Draw ray BL .

3. Complete the statement.

Since point L is in the interior of $\angle NBH$ then $m\angle NBL + m\angle LBH = m\angle NBH$.

Angles RKH , HKD , and DKT are shown where $m\angle RKH = (4y)^\circ$, $m\angle DKH = (6y - 16)^\circ$, $m\angle DKT = (2y + 4)^\circ$, and $m\angle RKT = 132^\circ$.



4. Determine the value of y .
 $4y + 6y - 16 + 2y + 4 = 132$
 $12y - 12 = 132$
 $12y = 144$
 $y = 12$

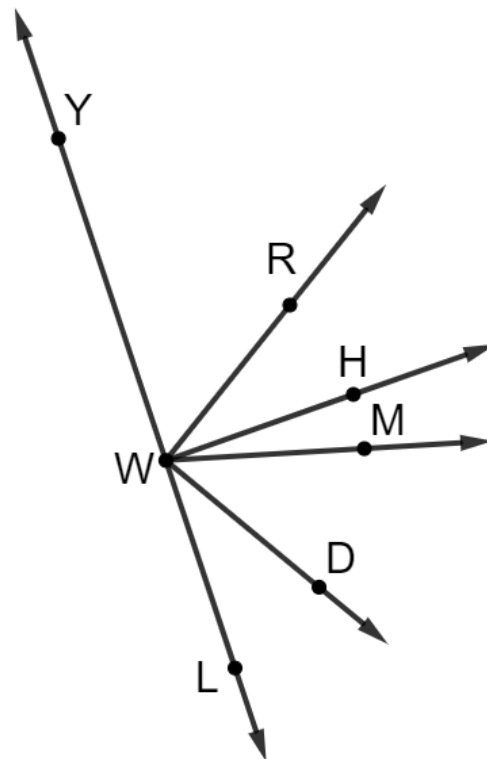
5. Determine $m\angle HKT$.
 $6y - 16 + 2y + 4$
 $8y - 12$
 $8 \cdot 12 - 12$
 84°

6. Determine $m\angle DKR$.
 $6y - 16 + 4y$
 $10y - 16$
 $10 \cdot 12 - 16$
 104°

Points R , H , M , and D lie in the interior of straight angle YWL such that $m\angle YWM = 105^\circ$, $m\angle MWD = 43^\circ$, $m\angle HWD = 59^\circ$, and $m\angle RWH = 2m\angle HWM$.

Determine each.

7. $m\angle YWD$
 $105^\circ + 43^\circ = 148^\circ$
8. $m\angle LWD$
 $180^\circ - 148^\circ = 32^\circ$
9. $m\angle HWM$
 $59^\circ - 43^\circ = 16^\circ$
10. $m\angle HWR$
 $2 \cdot 16^\circ = 32^\circ$



Geometry

Unit 1

Lesson 10 Homework

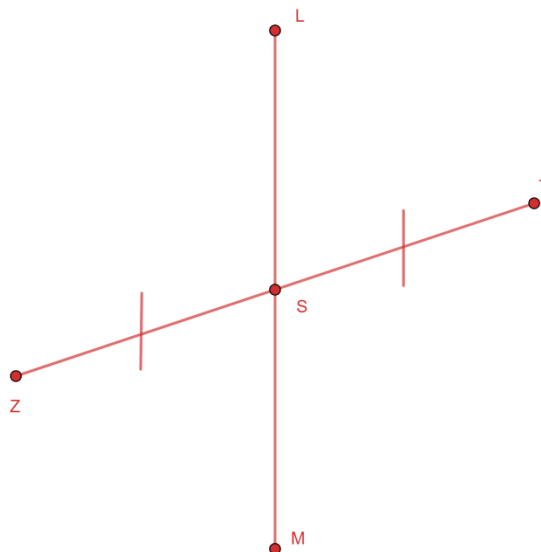
Name **Answer Key** _____

Date _____

Period _____

$\overline{LM}$ bisects $\overline{ZT}$ at point S . $\overline{ZS}$ has a length of $3x + 7$, and $\overline{ST}$ has a length of $-5x + 23$.

1. Draw a diagram of the description.



2. Find the value of x .

$$3x + 7 = -5x + 23$$

$$8x = 16$$

$$x = 2$$

3. Determine the length of $\overline{ZT}$.

$$\overline{ZT} = 3x + 7 + -5x + 23$$

$$= -2x + 30$$

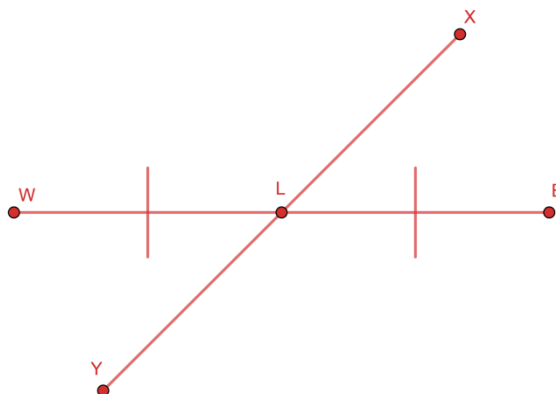
$$= -2(2) + 30$$

$$= -4 + 30$$

$$= 26$$

$\overline{XY}$ bisects $\overline{WB}$ at point L . $WB = 3x + 8$ and $WL = \frac{1}{2}x + 12$.

4. Draw a diagram of the description.



5. Find the value of x .

$$3x + 8 = 2\left(\frac{1}{2}x + 12\right)$$

$$3x + 8 = x + 24$$

$$2x = 16$$

$$x = 8$$

6. Find the length of $\overline{WB}$.

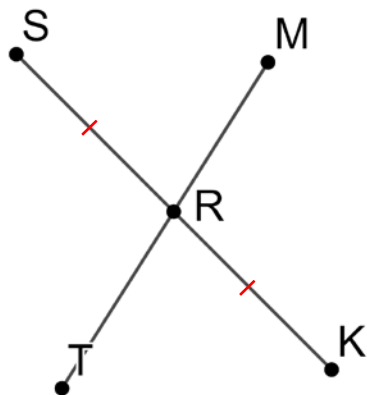
$$\overline{WB} = 3x + 8$$

$$= 3(8) + 8$$

$$= 24 + 8$$

$$= 32$$

$\overline{MT}$ bisects $\overline{SK}$ at point R as shown. $\overline{SK}$ has a length of $24x - 12$, $\overline{RK}$ has a length of $10x - 2$, $\overline{TR}$ has a length of $4x + 5$, and $\overline{RM}$ has a length of $8x - 5$.



7. Based on the given information which two segments are congruent?

$$\overline{SR} \cong \overline{RK}$$

8. Find the value of x .

$$24x - 12 = 2(10x - 2)$$

$$24x - 12 = 20x - 4$$

$$4x = 8$$

$$x = 2$$

9. Find the length of $\overline{RK}$.

$$\overline{RK} = 10x - 2$$

$$= 10(2) - 2$$

$$= 20 - 2$$

$$= 18$$

10. Find the length of $\overline{MT}$.

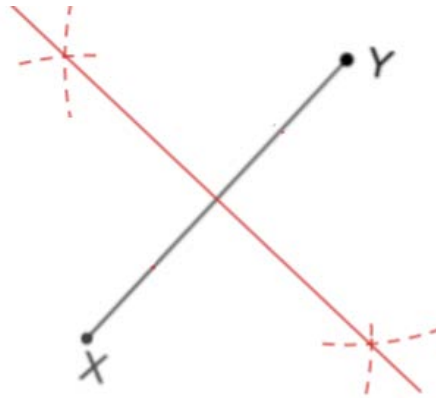
$$\overline{MT} = 4x + 5 + 8x - 5$$

$$= 12x$$

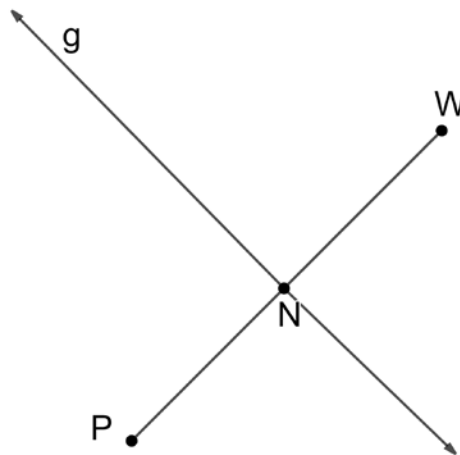
$$= 12(2)$$

$$= 24$$

1. Use a compass and a straightedge to construct a perpendicular bisector of $\overline{XY}$.



Line g is a perpendicular bisector of $\overline{PW}$ at point N .



2. If $PN = 4x + 17$ and $WN = 9x - 20$, find the value of x .

$$4x + 17 = 9x - 20$$

$$37 = 5x$$

$$x = \frac{37}{5} = 7.4$$

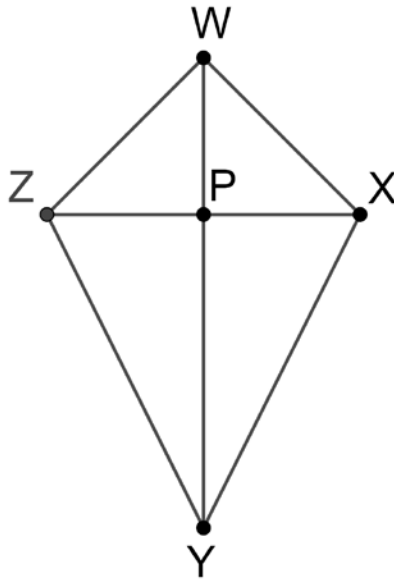
3. How long is $\overline{PN}$?

$$\overline{PN} = 4x + 17 = 4(7.4) + 17 = 46.6$$

4. How long is $\overline{PW}$?

$$\overline{PW} = \overline{PN} + \overline{WN} = 4x + 17 + 9x - 20 = 13x - 3 = 13(7.4) - 3 = 93.2$$

A quadrilateral is shown where $\overline{WY}$ is a perpendicular bisector of $\overline{ZX}$.



5. What angle is congruent to $\angle WZP$?
 $\angle WXP$
6. What angle is congruent to $\angle ZWP$?
 $\angle XWP$
7. Which angle(s) are congruent to $\angle WPZ$?
 $\angle WPX, \angle XPY, \angle ZPY$
8. What kind of triangle is $\triangle XYZ$?
Isosceles

$\overline{SL}$ is a perpendicular bisector of $\overline{RT}$, intersecting at point N . The length of $\overline{SR}$ is $\frac{2}{3}x - 40$ and the length of $\overline{ST}$ is $\frac{1}{4}x + 12$.

9. Find the value of x .

$$\frac{2}{3}x - 40 = \frac{1}{4}x + 12$$

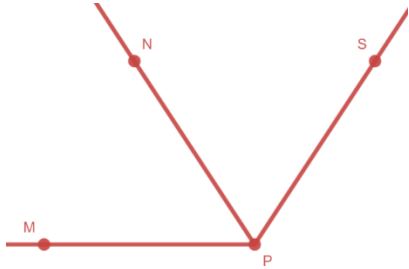
$$\frac{5}{12}x = 52$$

$$x = \frac{624}{5} = 124.8$$
10. Find the length of $\overline{ST}$.

$$\overline{ST} = \frac{1}{4}x + 12 = \frac{1}{4}(124.8) + 12 = 31.2 + 12 = 43.2$$

$\angle MPS$ is bisected by $\overrightarrow{PN}$. $m\angle MPN = (7x + 4)^\circ$ and $m\angle NPS = (16x - 68)^\circ$.

1. Draw a sketch that represents the description.



2. Find the value of x .

$$7x + 4 = 16x - 68$$

$$72 = 9x$$

$$x = 8$$

3. Determine $m\angle MPN$.

$$(7(8) + 4)^\circ = (56 + 4)^\circ = 60^\circ$$

4. Find $m\angle MPS$.

$$2(60^\circ) = 120^\circ$$

5. $\angle FBW$ is bisected by $\overrightarrow{BR}$ where the $m\angle FBW = (20x - 14)^\circ$ and $m\angle RBW = (\frac{5}{2}x + 8)^\circ$.

Find $m\angle FBR$.

$$20x - 14 = 2\left(\frac{5}{2}x + 8\right)$$

$$20x - 14 = 5x + 16$$

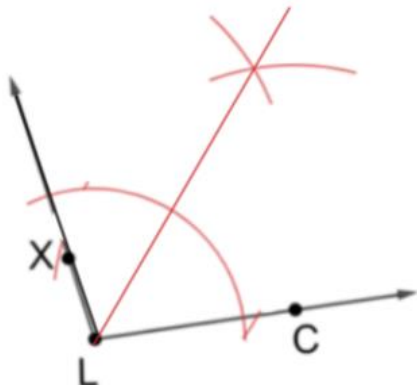
$$15x = 30$$

$$x = 2$$

$$\frac{5}{2}(2) + 8 = 5 + 8 = 13$$

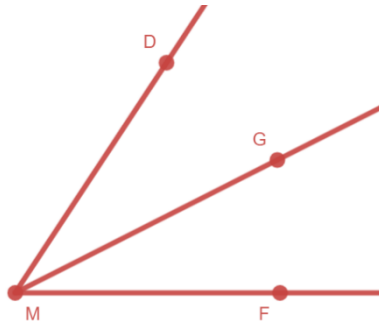
$$m\angle FBR = 13^\circ$$

6. Use a compass and a straightedge to bisect $\angle XLC$.



$\angle DMF$ is bisected by $\overrightarrow{MG}$ where $m\angle DMG = (4x)^\circ$ and $m\angle DMF = (2x + 24)^\circ$.

7. Sketch a diagram that could be used to represent this situation.



8. Find the value of x .

$$2x + 24 = 2(4x)$$

$$2x + 24 = 8x$$

$$24 = 6x$$

$$x = 4$$

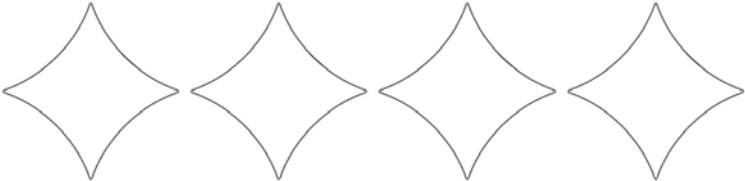
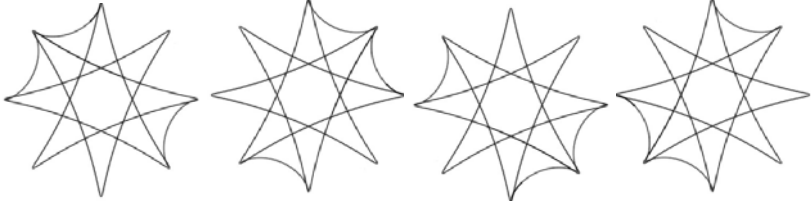
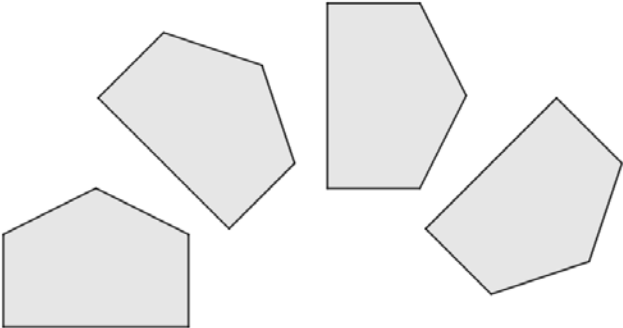
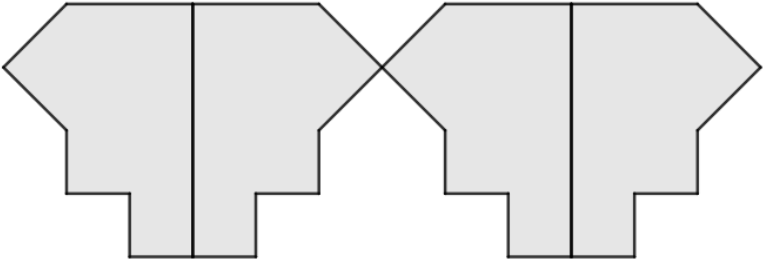
9. Find $m\angle DMG$.

$$(4(4))^\circ = 16^\circ$$

10. Find $m\angle DMF$.

$$(2(4) + 24)^\circ = (8 + 24)^\circ = 32^\circ$$

For each set of images name the type(s) of symmetry the image represents.

	Set of Four Images	Symmetry
1.		Translational, Reflectional, and Rotational
2.		Rotational
3.		Rotational
4.		Translational and Reflectional

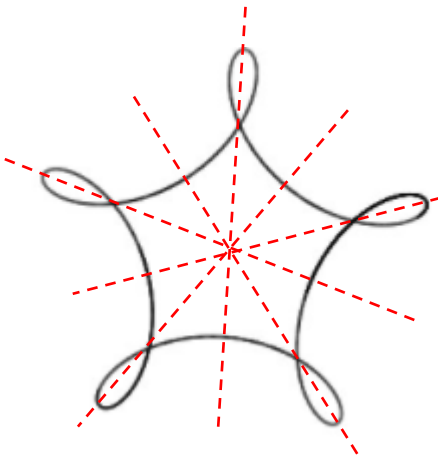
A figure is shown.

5. How many lines of symmetry does the figure have?

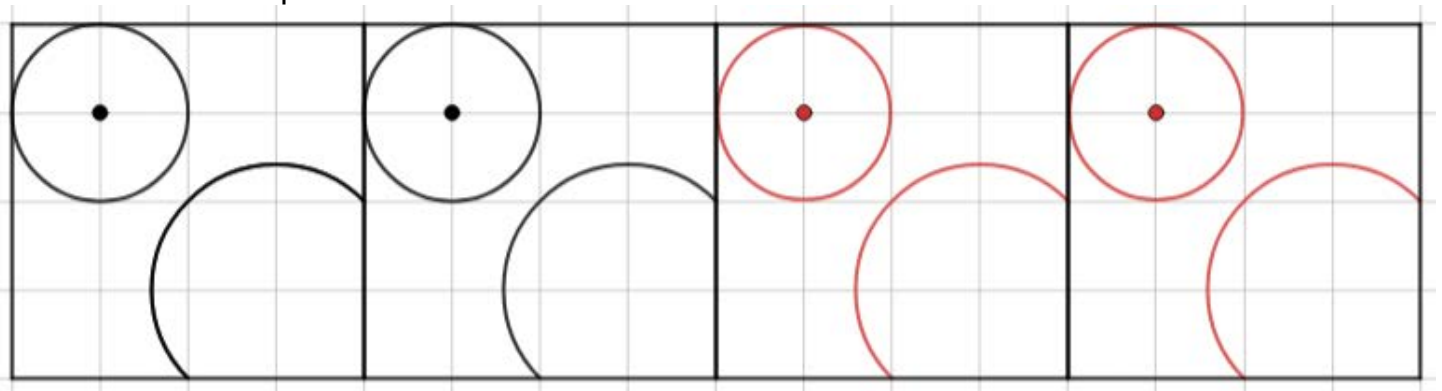
5

6. What is the order of rotational symmetry for the figure?

5



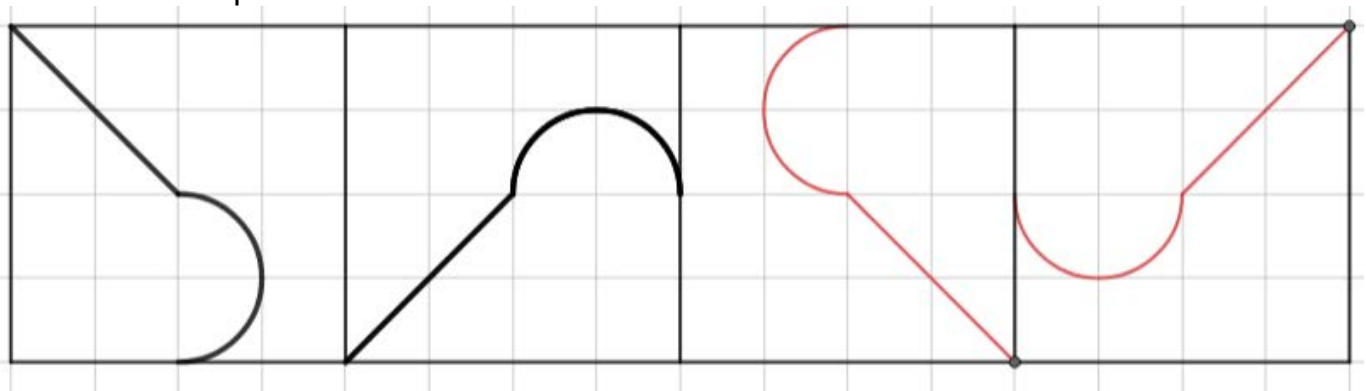
A series of four squares is shown.



7. Use the pattern in the first two squares to complete the pattern in the last two squares.
8. What type of symmetry is shown in this series of squares?

Translational symmetry

A series of 4 squares is shown.



9. Use the pattern in the first two squares to complete the pattern in the last two squares.
10. What type of symmetry is shown in this series of squares?

Rotational Symmetry